

APEX SYSTEMS

Engineering Tomorrow's Enterprise, Today

REQUEST FOR PROPOSAL — RESPONSE

Enterprise Digital Infrastructure Modernisation Programme (EDIM)

Reference: EDIM-2024-Q4-007

Submission Date: 15 November 2024 | Validity: 90 Days

YEARS ACTIVE	ENTERPRISE CLIENTS	CERTIFIED ENGINEERS	COUNTRIES SERVED
18	340+	520	32

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1. Executive Summary & Company Profile

Apex Systems Pte. Ltd. is an award-winning enterprise technology integrator headquartered in Singapore, with delivery centres in Bengaluru, Kuala Lumpur, Sydney, and Dubai. Founded in 2006, we have completed over 420 large-scale digital transformation programmes spanning banking, government, healthcare, and logistics. Our ISO 27001:2022, ISO 9001:2015, CMMI Level 5, and SOC 2 Type II certifications underpin an engineering culture of rigour, transparency, and client-centricity.

We are uniquely positioned to deliver the Enterprise Digital Infrastructure Modernisation Programme (EDIM) because we combine deep technical capability with regulated-industry compliance expertise and a proven Agile-at-Scale delivery model (SAFe 6.0). Our proposed solution is cloud-native, zero-trust by design, and built to exceed every functional and non-functional requirement stated in the RFP.

Attribute	Detail
Legal Name	Apex Systems Pte. Ltd.
Registration	200612345K — Accounting & Corporate Regulatory Authority, Singapore
Headquarters	18 Cross Street, #08-01, Singapore 048423
Delivery Centres	Bengaluru Kuala Lumpur Sydney Dubai
Founded	2006 (18 years of continuous operation)
Revenue FY2023	SGD 142 Million
Total Employees	1,850 globally
Certifications	ISO 27001:2022 ISO 9001:2015 CMMI Dev L5 SOC 2 Type II PCI-DSS v4
Cloud Partnerships	AWS Premier Partner Microsoft Azure Expert MSP GCP Authorized Partner
Insurance	Professional Indemnity SGD 10M Public Liability SGD 5M Cyber SGD 5M

2. Understanding of Requirements

After reviewing the full RFP package, three annexures, and clarification notes issued 01 Nov 2024, Apex Systems confirms understanding of six core requirement themes. All are fully addressed within this proposal:

#	Requirement Theme	Our Interpretation	Status
R-01	Legacy system decommission	Migrate 6 on-premise legacy apps — cloud-native, zero data loss	Fully Addressed
R-02	Real-time data integration	Event-driven integration hub @ 50,000 TPS peak throughput	Fully Addressed
R-03	Security posture uplift	Zero-trust, MFA, SIEM, EDR across 2,400 endpoints	Fully Addressed
R-04	Disaster recovery	RPO ≤ 15 min RTO ≤ 1 hour for Tier-1 workloads	Fully Addressed
R-05	Regulatory compliance	MAS TRM 2021, PDPA, ISO 27001 within programme timeline	Fully Addressed
R-06	User adoption & training	800 staff 3 role tracks multi-modal LMS delivery	Fully Addressed

3. Proposed Technical Solution

3.1 Architecture Overview

Our solution follows a cloud-native micro-services architecture deployed on AWS (primary) with Azure as the cross-region DR target. The stack is structured across six horizontal layers:

Layer	Components	Technology Stack	Redundancy
Presentation	Web portal, Mobile (iOS/Android)	React 18, Flutter 3.16, AWS API Gateway	Multi-AZ + CloudFront CDN
Application	10 domain micro-services (Auth, User, Item, Cart, Order, Shipping, Billing, Profile, Analytics, Admin)	Node.js 20, Python 3.11, Go 1.21, AWS Lambda	Multi-AZ + ElastiCache
Integration	Event-driven bus, REST/GraphQL APIs	Apache Kafka, AWS EventBridge	High Availability factor 3
Data	Transactional DB, DW, Object Storage	PostgreSQL 15, Amazon Redshift, S3, Redis 7.2	Multi-AZ + S3 CRR
Security	WAF, SIEM, Secrets Mgr, KMS	AWS WAF, Wazuh, HashiCorp Vault	Active AWS IAM, SSM, S3
Observability	APM, distributed tracing, log aggregation	Datadog, Jaeger, Fluent Bit, CloudWatch	Managed SaaS

3.2 Performance & Scalability Benchmarks

Metric	Target (RFP)	Apex Design Capacity	Testing Method
API Throughput	10,000 TPS	55,000 TPS (5.5× headroom)	Gatling 72-hr soak test
API P95 Latency	< 300 ms	< 95 ms at 50K TPS	Percentile profiling under load
DB Query P99	< 500 ms	< 120 ms	pgBench + custom workload
Batch Processing	500K records/hr	2.1M records/hr	AWS Glue + Spark profiling
UI Load Time (LCP)	< 3 s	< 1.1 s	Lighthouse CI in pipeline
Availability	99.9% (43.8 min/mo)	99.95% design target	Multi-AZ + health checks
RPO	≤ 15 min	≤ 5 min (continuous replication)	AWS DMS continuous
RTO	≤ 1 hr	≤ 18 min automated failover	Chaos engineering quarterly

4. Implementation Plan & Milestones

The programme is structured across four phases spanning 18 months using SAFe 6.0 Agile-at-Scale. Each phase concludes with a client-witnessed acceptance gate before the next commences. Total elapsed duration: 72 weeks (Discovery: 3 wks | Build: 10 wks per wave | Testing: 4 wks | Go-live: 2 wks per wave).

Phase	Weeks	Key Deliverables	Exit Criteria
Phase 0 Mobilisation	1–4	Project charter; RACI; environment provisioning; risk register	Environment classification onboarded; charter signed
Phase 1 Foundation	5–20	Cloud landing zone; IAM; CI/CD; core data platform; integrations	Security baseline complete; iHub handles 10K Tbps
Phase 2 Migration	21–44	6 legacy apps migrated; 10 micro-services deployed; full + UAT	Data sign-off; migration UATs closed; 3 benchmark tests passed
Phase 3 Compliance	45–56	Pen-test remediation; MAS TRM artefacts; PDPA assessments	ISO 27001 Stage 2 passed; DR RTO < 18 min
Phase 4 Go-Live	57–72	3-wave go-live; hypercare support; legacy decommission; ZK PoA	Go-live on day 1; hypercare; client acceptance

Risk Register — Top 8

ID	Risk	Likelihood	Impact	Mitigation
R01	Legacy data quality causing migration delays	Medium	High	Data profiling sprint in Phase 0; automated DQ rules
R02	Third-party API instability	Medium	Medium	Circuit breakers; shadow mode testing; SLA in vendor contracts
R03	Key personnel unavailability	Low	High	Shadow resources named; knowledge-transfer protocol
R04	Regulatory requirement changes (MAS)	Medium	High	Dedicated compliance liaison; bi-weekly regulatory scan
R05	Cloud cost overrun	Medium	Medium	Cost Anomaly Detection; weekly FinOps review; reserved instances
R06	User adoption resistance	Medium	Medium	Change management track; champion network; phased roll-out
R07	Scope creep from stakeholders	High	High	Formal change control board; impact analysis within 5 days
R08	CBS integration takes longer than anticipated	Medium	High	CBS in Phase 1 with 4-week buffer; dedicated SME liaison

5. Team Structure & Key Personnel

A dedicated 34-person team will be exclusively allocated to EDIM: 12 Singapore-based client-facing and 22 offshore (Bengaluru). All personnel hold current business/work authorisations.

Role	Name	Location	Experience	Key Certifications
Programme Director	Ms. Priya Subramaniam	Singapore	20 yrs	PMP SAFe SA TOGAF 10
Technical Architect	Mr. Rajan Krishnamurthy	Singapore	17 yrs	AWS SA Pro CKAD TOGAF
Security Lead	Mr. David Lim Eng Kiat	Singapore	15 yrs	CISSP CISM ISO 27001 LA
Cloud Infra Lead	Ms. Ananya Menon	Bengaluru	13 yrs	AWS DevOps Pro Terraform
Integration Lead	Mr. Sriram Venkataraman	Singapore	14 yrs	MuleSoft Certified Kafka
Data Engineering Lead	Ms. Farah Lim	Singapore	12 yrs	Databricks dbt Certified
QA Lead	Mr. Karthik Naidu	Bengaluru	11 yrs	ISTQB Expert Gatling
Change Mgmt Lead	Ms. Grace Ong Hui Lin	Singapore	16 yrs	Prosci ADKAR PMP
Senior Developers (×8)	Bengaluru Pod A	Bengaluru	6–10 yrs	AWS Developer Azure Fundamentals
DevOps Engineers (×4)	Bengaluru Pod B	Bengaluru	5–8 yrs	AWS DevOps Kubernetes CKAD
QA Engineers (×4)	Bengaluru Pod C	Bengaluru	4–7 yrs	ISTQB Foundation Selenium
Business Analysts (×3)	Singapore BA Team	Singapore	6–9 yrs	CBAP Agile BA

6. Commercial Proposal & Pricing

All prices are in Singapore Dollars (SGD), exclusive of 9% GST. Phases 0–3 are fixed-fee; Phase 4 hypercare is capped T&M. Payment schedule: 20% on award | 20% Phase 1 | 25% Phase 2 UAT | 25% Phase 3 audit | 10% final acceptance.

Line Item	Phase	Qty	Unit Rate (SGD)	Total (SGD)
Programme Management & Governance	All	18 mths	28,000/mth	504,000
Cloud Landing Zone & Infrastructure	Phase 1	Fixed	380,000	380,000
Integration Hub Development	Phase 1	Fixed	290,000	290,000
Core Data Platform (DW + Lakehouse)	Phase 1	Fixed	320,000	320,000
Legacy App Migration (×6)	Phase 2	6 apps	140,000/app	840,000
New Micro-services (×10)	Phase 2	10 svcs	85,000/svc	850,000
Security Hardening & Compliance	Phase 3	Fixed	280,000	280,000
User Training (800 users, 3 tracks)	Phase 3	Fixed	95,000	95,000
Go-Live Support & Hypercare (3 waves)	Phase 4	3 mths	55,000/mth	165,000
Documentation & Knowledge Transfer	Phase 4	Fixed	45,000	45,000
			Sub-Total (SGD)	3,769,000
			Contingency Reserve (5%)	188,450
			TOTAL FIXED PRICE (excl. GST)	3,957,450

Cloud consumption (AWS/Azure) is pass-through at list price — estimated SGD 18,000–22,000/month post go-live. Travel capped at SGD 40,000 across the programme.

7. Security, Compliance & Risk Controls

Security is embedded via a DevSecOps (Shift-Left) model. Every commit is SAST/DAST scanned. IaC templates are validated against CIS Benchmarks before deployment. Key controls:

Domain	Controls Implemented	Framework Alignment	Validation
Identity & Access	ZTNA; MFA enforced; PAM for privileged accounts	NIST SP 800-207, ISO 27001, MAS TRM 2022	Quarterly access review; PAM session recording
Data Protection	AES-256 at rest; TLS 1.3 in transit; field-level encryption; DPA; PIA	GDPR, HIPAA, PCI-DSS, MAS TRM 2022	Annual DLP audit; automated PII scan in pipeline
Threat Detection	Wazuh SIEM; 200+ detection rules; 24/7 SOC; EDR on all endpoints	NIST CSF, MITRE ATT&CK, MAS TRM 2022	Monthly threat hunt; SOC KPI dashboard
Vulnerability Mgmt	Monthly VAPT; continuous SAST/DAST in CI/CD; CVE tracking	CWE Top 25, CVEs, MAS TRM 2022	Quarterly pen-test report
Business Continuity	Multi-AZ primary; Azure cross-region DR; 2001 RTO; MAS TRM 2022	ISO 22301, MAS TRM 2022	Bi-annual DR drill report
Audit & Logging	Immutable logs (CloudTrail + S3 Object Lock); 7-year retention	MAS TRM 2022, PIA	Log integrity check monthly
Supply Chain Security	Vendor assessment; SBOM for all OSS; Signed binaries; SLA review	SAS 9001, MAS TRM 2022	Annual vendor review; SBOM in CI/CD

Certification Portfolio

Certification	Issuing Body	Status	Next Audit
ISO 27001:2022 — Information Security	BSI Group	CERTIFIED	Oct 2026
ISO 9001:2015 — Quality Management	TÜV SÜD	CERTIFIED	Mar 2026
SOC 2 Type II — Security & Availability	KPMG LLP	CERTIFIED (report available)	Jul 2025
CMMI Dev Level 5 — Process Maturity	ISACA / CMMI Institute	CERTIFIED	Aug 2026
PCI-DSS v4.0	Foresite Cybersecurity (QSA)	COMPLIANT	Dec 2025
MAS TRM Self-Assessment	Internal (MAS framework)	COMPLETED — submitted	Annual

8. Support Model & SLA Framework

Priority	Definition	Initial Response	Resolution	Escalation
P1 – Critical	Production down; >50% users impacted; data loss	15 mins (24x7)	4 hrs	Auto-escalate to CTO at 2 hr
P2 – High	Major feature impaired; workarounds possible	1 hr (24x7)	8 business hrs	Account Director at 6 hr
P3 – Medium	Non-critical; moderate impact	4 business hrs	3 business days	Delivery Manager at 2 days
P4 – Low	Cosmetic / informational	1 business day	10 business days	Standard ticket

Post go-live AMS team: 2 app support engineers, 1 cloud infrastructure engineer, 1 part-time security analyst (on-call). Support channels: JIRA Service Management portal, Slack Connect, 24x7 emergency hotline.

9. Relevant Experience & Client References

Client	Sector	Project Summary	Value	Duration	Outcome
MegaBank Asia (SG)	Financial	Core Banking Modernisation — 12 legacy systems to 2 cloud SAP integrations	\$5.6M	24 months	Zero data loss; 99.97% uptime Yr 1;
National Healthcare Group	Healthcare	Clinical data platform; real-time analytics, GDPR compliance	\$3.8M	18 months	40% faster reporting; ISO 27001 cert
Singapore Land Authority	Government	GIS digital twin; integrated with 6 gov data sources, 25m+ data	\$2.1M	23 months	8 weeks ahead of schedule; GovTech
LogiFreight Holdings (MY)	Logistics	Supply chain visibility; real-time IoT; ML-based ETA predictions	\$6.2M	15 months	35% improvement in on-time delivery
OmniRetail Group (AU)	Retail	Omnichannel e-commerce; Kafka inventory sync; PCI-DSS	\$4.5M	20 months	3x peak load on Black Friday; 0 P1 in

Reference contacts are available upon request. Apex Systems consents to the evaluation team contacting all named references during the evaluation period.

10. Assumptions & Exclusions

- Client will provide dedicated SME access within 5 business days of a formal request.
- AWS master payer account provisioned by client within 2 weeks of contract award.
- Third-party software licences not listed are the client's responsibility.
- Scope is limited to 6 legacy apps in Annexure B. Additional apps are change requests.
- Data migration covers production data only. Archives beyond 7 years are excluded.
- DR testing is conducted outside peak hours (06:00–08:00 SGT Sundays).
- MAS regulatory submissions are prepared by Apex Systems and lodged by the client's compliance officer.

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Date: 15 November 2024

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